

Polynomial curves

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- Polynomial curves and curves with one place at infinity.
- Invariants.
- Embedding of polynomial curves in the plane.
- Cuspidal polynomial curves.
- Classification of polynomial curves with small degrees.

Notations:

- $\mathbb{K}$ is an algebraically closed field of characteristic zero, and $x(t), y(t) \in \mathbb{K}[t] \setminus \mathbb{K}$, $\deg_t x(t) = n > m = \deg_t y(t)$.
- $V = \{(x(a), y(a)) \in \mathbb{K}^2 \mid a \in \mathbb{K}\}$ is the algebraic curve defined by $x(t), y(t)$: V is a polynomial curve.

Assertion. The implicit function of V is of the form $f(x, y) = y^n + \sum_{i=1}^n a_i(x)y^{n-i}$, and f has one point at infinity, $y^n = 0$. We write $f_\infty(u, y) \in \mathbb{K}[[u, y]]$ for the localization of h_f at the point at infinity. Then $f_\infty(u, y)$ is irreducible. We say that f has one place at infinity.

Example. $x(t) = t^6 + t, y(t) = t^4$ gives $f = (y^3 - x^2)^2 - 4xy^2 - y$,

Semigroup. $\Gamma(f) = \{\text{int}(f, g) = \deg_t g(x(t), y(t)) \mid f \nmid g\}$

Proposition. $\Gamma(f)$ is a free numerical semigroup.

Proposition. We can calculate a system of generators of $\Gamma(f)$ only from the equation, using the theory of approximate roots.

Definition. $d \mid n$, g a monic polynomial of degree n/d . We say the g is the d -th approximate root of f if in the expansion of f w.r.t. g :

$$f = g^d + \alpha_1(x, y)g^{d-1} + \cdots + \alpha_d(x, y),$$

$\alpha_1(x, y) \neq 0$. A d -th approximate root of f exists and it is unique. We denote it $\text{App}(f, d)$.

In the example. $f = (y^3 - x^2)^2 - 4xy^2 - y$: $r_0 = d_1 = 6$, $r_1 = 4$, $d_2 = \gcd(4, 6) = 2$, then $g = \text{App}(f, 2) = y^3 - x^2$, and $r_2 = \text{int}(f, g) = \text{rank}_{\mathbb{K}} \mathbb{K}[x, y]/(f, g) = 7$. As $\gcd(d_2, r_2) = d_3 = 1$, we get $\Gamma(f) = \langle 4, 6, 7 \rangle$.

Invariants:

- $\mu = \text{int}(f_x, f_y) = \text{rank}_{\mathbb{K}} \mathbb{K}[x, y]/(f_x, f_y)$: the global Milnor number of the family $(f - \lambda)_\lambda$.
- $\mu(f)$ is the global Milnor number of the curve $V(f)$, equal $\text{rank}_{\mathbb{K}}(\mathbb{K}[x, y]/(f_x, f_y))_{(f)}$
- $\nu(f) = \text{rank}_{\mathbb{K}} \mathbb{K}[x, y]/(f, f_x, f_y)$

Singularities:

- Cusps: $x'(t) = y'(t) = 0$.
- Double points: $\frac{x(t) - x(s)}{t - s} = \frac{y(t) - y(s)}{t - s} = 0$.

Clearly $\nu := \nu(f) \leq \mu(f) \leq \mu$.

Proposition (–A. Sathaye) $\nu \geq \frac{\mu}{2}$. In particular $\nu \in [[\mu/2, \mu]]$.

$\mu/2 - - - - - \mu$.

A first result

Abhyankar-Moh Theorem. Suppose that $\mu(f) = 0$ (in particular $\nu = 0$, but also $\mu = 0$). Then $V(f)$ is equivalent to a coordinate, i.e. there exists an automorphism σ of $\mathbb{K}[x, y]$ such that $\sigma(f) = y$.

Variants.

- μ is the conductor of $\Gamma(f)$. If $\mu(f) = 0$ then $\Gamma(f) = \mathbb{N}$.
- $\mathbb{K}[x(t), y(t)] = \mathbb{K}[t]$ implies that $m|n$.
- A rational non singular curve with one place at infinity is equivalent to a coordinate.

Example. $x(t) = t^4 - t, y(t) = t^2$. We have $f = (y^2 - x)^2 - y$. We have $\mathbb{K}[f, y^2 - x] = \mathbb{K}[y^2 - x, y] = \mathbb{K}[y, x]$.

Definition. Let $\mathcal{A} = \text{Aut}(\mathbb{K}[x, y])$ be the set of automorphisms of $\mathbb{K}[x, y]$. We say that g is **equivalent** to h ($g \sim h$ if $h = \sigma(g)$ for some $\sigma \in \mathcal{A}$). We say that $V(g)$ is **isomorphic** to $V(h)$ ($V(g) \simeq V(h)$) if their ring of coordinates are isomorphic.

Isomorphism and equivalence

Embedding of curves: If $g \sim h$ then $V(g) \simeq V(h)$. The converse is not true. If $V(g) \simeq V(h)$ but $g \not\sim h$, then we say that $V(g)$ has two non equivalent emddings in the plane.

Example: A coordinate has a unique embedding in the plane.

Example. Hyperbola. $g_k = 1 - xy^k, k \geq 1$. Then $V(g_k)$ is parametrized by $x = t^{-k}, y = t$. Take

$$\phi : V(g_1) \mapsto V(g_k), \phi(a, b) = (a^k, b)$$

and

$$\psi : V(g_k) \mapsto V(g_1), \psi(c, d) = (cd^{k-1}, d).$$

Then $\phi \circ \psi = \text{Id}$, and $\psi \circ \phi = \text{Id}$, but $g_1 \not\sim g_k, k \geq 2$: The hyperbola has infinitely many embeddings in the plane (but the **parabola** has a unique embedding in the plane).

Embedding of polynomial curves

Conjecture (Shpilrain, Yu): A polynomial curve has a unique embedding in the plane.

A counter example:

$x = t^4 - t, z = t^3 - 1, y = z^2 + \frac{z}{2} = t^6 - (3/2)t^3 + (1/2)$. We have $\mathbb{K}[x, y] = \mathbb{K}[x, z]$, so $V(g) : (x(t), y(t)) \simeq V(h) : (x(t), z(t))$.

Proposition. $g \not\sim h$.

Proof. $\Gamma(h) = \langle 4, 3 \rangle$, and $\Gamma(g) = \langle 6, 4, 3 \rangle$. **This finishes the proof? see later.**

Definition. r-sequence. A sequence $(r_0, r_1, \dots, r_h)$ is called an r-sequence if the following conditions hold.

- $r_0 > r_1$ and r_1 does not divide r_0 .
- $\Gamma = \langle r_0, \dots, r_h \rangle$ is a free numerical semigroup for the above arrangement.
- If $d_k = \gcd(r_0, \dots, r_{k-1})$, then $r_k d_k > r_{k-1} d_{k-1}$ for all $k \geq 2$.

The $\underline{r}$ -sequence

Fact. We can associate with any curve with one place at infinity a unique $\underline{r}$ -sequence.

Examples. 1. $g = (y^2 - x)^3 - y$: $r_0 = 6, r_1 = 3, r_2 = 1$:
 $Y = y^2 - x, X = y$, we get $g = Y^3 - X$, then a coordinate, see y .
The $\underline{r}$ -sequence is 1.

2. $g = (y^2 - x)^2 - y^3$: $r_0 = 4, r_1 = 2, r_2 = 3$. We set
 $Y = y^2 - x, X = y$, we get: $g = Y^2 - X^3$, then we interchange X
and Y , The $\underline{r}$ -sequence is 3, 2.

Remark. (—, Pedro)

- Given $g \in \mathbb{K}[x, y]$, we can decide if g has one place at infinity. with one place at infinity (Abhyankar). The algorithm is implemented in GAP.
- If g has one place at infinity, then we can construct $h = y^n + a_2(x)y^{n-2} + \dots + a_n(x)$ which is equivalent to g , such that $r_0 = n > r_1$, and r_1 does not divide r_0 . The algorithm is implemented in GAP.

The $\underline{r}$ -sequence

Proposition (Abhyankar-Sathaye) Let f, g be two polynomials with one place at infinity. If $f \sim g$ then f and g have the same $\underline{r}$ -sequence.

Back to polynomial curves. Here are two questions raised by Abhyankar: let $\underline{r} = (r_0, \dots, r_h)$ be an $\underline{r}$ -sequence, and let Γ be the numerical semigroup generated by $\underline{r}$.

- Is $\underline{r}$ the $\underline{r}$ -sequence of a polynomial curve?
- Is Γ the semigroup of a polynomial curve?

The answer to the two questions is no (Sathaye-Stenerson).

1. The $\underline{r}$ -sequence $(8, 6, 3)$ cannot be associated with a polynomial curve (however, it is associated with $(y^4 - x^3)^2 - y$).
2. The numerical semigroup $\langle 22, 6, 17 \rangle$ is not the semigroup of a polynomial curve (proposed by S-S and proved by Fujimoto, Suzuki, Yokoyama).

External polynomial curves

The $\underline{r}$ sequence $(8, 6, 3)$. It is associated with of the one place polynomial $g = (y^4 - x^3)^2 - y$. It is also the semigroup of any $g - \lambda, \lambda \in \mathbb{K}$. In particular, the family $(g - \lambda)_{\lambda \in \mathbb{K}}$ does not contain rational elements.

Question. What is the maximum number of rational elements in a family $(g - \lambda)_{\lambda}$?

Theorem (-). A family $(g - \lambda)_{\lambda}$ of curves with one place at infinity contains at most two rational elements. Moreover, if the family contains exactly two rational elements, say g_1 and g_2 , then the following holds.

- $\mu(g_1) = \mu(g_2) = \mu/2$.
- Singularities of g_i are all double points, $i = 1, 2$ (with local Milnor numbers= 1).

Extremal Polynomial curves

f is a polynomial curve. Recall that $\nu \in [[\mu/2, \mu]]$. If $\nu = \mu/2$, then all singularities of $V(f)$ are double points. What if $\nu = \mu$?

Theorem. (– Avinash Sathaye, –Pedro) If $\mu = \nu$ then f is equivalent to a quasi-homogeneous polynomial ($Y^a - X^b$).

Idea of the proof: Imitate Zariski proof in the local case: $\nu = \mu$ implies that every differential $w = adx + bdy$ on the curve is exact ($= dg$). If f is not equivalent to a quasi-homogeneous polynomial, then, after a change of variables, $w = nx dy - my dx$ is not exact.

Corollary. If $f \in (f_x, f_y)$, then $f \sim Y^a - X^b$.

Not easy to characterize polynomials g such that $g \in (g_x, g_y)$. For example, for which g do we have $(g_x, g_y) = (1)$?

Now we know how to characterize polynomial curves for which $\nu = \mu$ (resp. $\nu = \mu/2$). But we have more...

Lin-Zaidenberg Theorem

Let g be a polynomial of $\mathbb{K}[x, y]$. If $p \in V(g)$, we say that $V(g)$ is unibranch at p if the local equation of $V(g)$ at m_p is irreducible (if $p = (0, 0)$, then this means that g is irreducible in $\mathbb{K}[[x, y]]$).

Theorem (V. Lin-M. Zaidenberg) Suppose that g has one place at infinity. If $V(g)$ is unibranch at finite distance, then $g \sim Y^b - X^a$

Remarks.

- Link with polynomial curves: we can prove that under the hypotheses of the theorem, g is a polynomial curve.
- More general than Abhyankar-Moh Theorem: if $V(g)$ is not singular at finite distance, then it is unibranch, so $g \sim$ to a coordinate (either $a = 1$ or $b = 1$).
- Problem: The proof uses the geometry of surfaces. It does not work on an arbitrary field.

Question Is there an algebraic proof of L.-Z. Theorem?

Lin-Zaidenberg Theorem

Remarks. 1. Suppose that $V(g)$ is singular at finite distance. Unibranch means that all singularities are cusps. So we have a global semigroup $\Gamma(g)$, and local semigroups, one at each cusp.

Question. Is it possible to use these structures, the theory of approximate roots.. in order to find an algebraic proof of L-Z theorem?

2. L.-Z. Theorem $\implies$ if all singularities of a polynomial curve are cusps, then we have only one cusp at finite distance. This is equivalent to say that if we have two cusps at finite distance, then we have at least one double point: the system

$$G(t, s) = \frac{x(t) - x(s)}{t - s} = 0, H(t, s) = \frac{y(t) - y(s)}{t - s} = 0$$

has at least one solution.

3. Another equivalent statement: if all singularities of a polynomial curve are cusps, then $\nu = \mu$.

Lin-Zaidenberg Theorem

Theorem (—Mahboub, work in progress) Let g be a polynomial curve.

- If $V(g)$ has at least two cusps, then $V(g)$ has at least one double point.
- If $V(g)$ has only one cusp, then it is equivalent to a quasi-homogeneous polynomial.

Idea of the proof: In the second point we used the arithmetic of the local and global semigroups. In the second point we studied the factorization of G, H as polynomials in $\mathbb{K}(t)[s]$, and how do they intersect:

$$G(t, s) = \prod_{i=1}^{n-1} (t - t_i(s)) \quad \text{and} \quad H(t, s) = \prod_{j=1}^{m-1} (t - \bar{t}_j(s))$$

Obviously this is related to the theory of resultants. In this case, the resultant in $t, \prod_{i,j} (t_i(s) - \bar{t}_j(s))$ is called Taylor-Abhyankar-resultant. It has been proved by Abhyankar that this resultant is of the form

$$r(s) = \prod_{k=1}^p (s - a_k)^{e_k}$$

with degree equal to the conductor of the semigroup ($= \mu$) and multiplicities equal to the orders of the conductor ideals.

Example. $x(t) = t^4 + t^2$, $y(t) = 2t^3 + t$:

$$r(s) = 8(s - i/\sqrt{2})^3(s + i/\sqrt{2})^3$$

.

Lin-Zaidenberg $\implies$ If $e_i = \mu_i$ for all i , then $p = 1$.

When $\mu - \nu$ is small

f is a polynomial curve: If $\nu = \mu$ then f is equivalent to a coordinate. If $\nu = \mu/2$ then f has only double points (with local Milnor numbers = 1). What if $\mu/2 < \nu < \mu$?

- $w = a(x, y)dx + b(x, y)dy$, we can associate its degree, which is the degree in t of $a(x(t), y(t))x'(t) + b(x(t), y(t))y'(t)$.
- If $\mathbf{A} = \mathbb{K}[x(t), y(t)]$, then this suggests to consider the module $\mathbf{M} = x'(t)A + y'(t)A$. Then we have the semigroup $\Gamma(f)$, and $d(\mathbf{M}) = \{\deg(w) | w \in \mathbf{M}\}$ (ideal of the semigroup). Inside $d(\mathbf{M})$ we have the degrees of exact differentials (degrees of $(g(x(t), y(t)))', g \in \mathbf{A}$), hence $\{a - 1 | a \in \Gamma(\mathbf{f})\}$, but we probably have some more elements: non exact degrees.

Proposition $\mu - \nu$ is nothing but the cardinality of non exact differentials.

When $\mu - \nu$ is small

Proposition. (...Pedro-Abbas):

- An algorithm for the calculation of $d(\mathbf{M})$ (then for the calculation of ne = non exact degrees).
- If $(r_0, r_1, \dots, r_h)$ is the $\underline{r}$ -sequence of f , then $ne \geq 2^{h-1}$, whence:

In particular

- If $ne = 1$, then $h = 1$. Moreover, $(r_0, r_1) = (2, 2p + 1), (4, 3), (5, 3)$.
- If $ne = 2$, then $h = 1, 2$. Moreover, if $h = 1$, then $(r_0, r_1) = (2, 2p + 1), (3, n), (4, 5), (4, 7) \dots$

Question. Is it possible to have a classification in terms of equations?

Answer: yes, but... There is a lot of calculations.

When $\mu - \nu$ is small

–, M. Hamie and Allan steel (Sydney): Classification of polynomial curves with small degrees. Work in progress. Suppose that the degree is ≤ 5 (then $\mu \leq 8$), then it is possible to get the possible equations as well as the description of their singularities.

Strategy: write the parametric equations (for example, $x(t) = t^4 + at^2 + bt$, $y(t) = t^3 + ct^2 + dt$), and apply the algorithm of calculation of $d(\mathbf{M})$. This way we get a big system of equations in the parameters (a, b, c, d in the example), then take the radical of the system, followed by the primary decomposition. This gives us a small system of equations, then the possible equations with the stratification in terms of ν .

Remark. This includes the case $ne = 1$, some subcases when $ne = 2$.

$ne = 1$: we get the following families: $x(t) = t^4 + at^2 + bt$, $y(t) = t^3 + ct$, with (a, b, c) is one of the following:

- $(k, \pm(4/3)\sqrt{2/3}k^{3/2}, (-1/2)k)$.
- $(k, \pm(2/3)\sqrt{1/3}\sqrt{-k}k, k)$.
- $(k, 0, (1/2)k)$.
- $(k, 0, (3/2)k)$.

Another example. $x(t) = t^5 + kt^2$, $y(t) = t^3$, $k \neq 0$: $ne = 2$.

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